AMATH 582: HOMEWORK 3

CALEB CRAMER

Applied Mathematics, University of Washington, Seattle, WA
cbcramer@uw.edu

Abstract. This report investigates the implementation and performance of Randomized Singular Value Decomposition (rSVD) as a computationally efficient alternative to deterministic Full SVD for dimensionality reduction. We explore the theoretical foundations of randomized matrix approximations, including Gaussian sampling and error bounds derived from concentration of measure phenomena. The algorithms are applied to a set of random functions and video datasets to perform background subtraction. Computational results demonstrate that rSVD achieves near-optimal low-rank approximations with significantly reduced computational complexity, particularly when using appropriate oversampling parameters.

1. Introduction and Overview

Singular Value Decomposition (SVD) and Principal Component Analysis (PCA) are fundamental tools in data science for dimensionality reduction, noise reduction, and feature extraction. However, as modern datasets grow in size, the computational cost of deterministic SVD algorithms—typically $O(mn^2)$—becomes prohibitive. This report explores Randomized SVD (rSVD), a probabilistic algorithm that leverages random projections to identify the subspace capturing the majority of a matrix's action. By constructing a smaller sketch of the input matrix, we can compute an approximate factorization much faster than standard methods while maintaining high accuracy. We implement the rSVD algorithm, verify theoretical error bounds against a baseline of random functions, and apply the method to a video background removal task (separating static background from moving foreground). The results validate the efficacy of randomized methods for high-dimensional data processing.

2. Theoretical Background

2.1. **QR Decomposition.** The QR decomposition factorizes a matrix $A \in \mathbb{R}^{m\times n}$ into an orthogonal matrix Q and an upper triangular matrix R [4]. The orthonormality of Q (where $Q^\top Q = I$) is guaranteed by the construction process, typically via the Gram-Schmidt process or Householder reflections. In the context of randomized PCA, Q serves as an orthonormal basis for the subspace that captures the action of A. The process ensures Q is orthonormal because each successive column is projected onto the orthogonal complement of the span of previous columns and then normalized:

$$q_j = \frac{a_j - \sum_{i=1}^{j-1}\langle q_i, a_j\rangle q_i}{\|a_j - \sum_{i=1}^{j-1}\langle q_i, a_j\rangle q_i\|}$$

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2.2. **Singular Value Decomposition.** The Singular Value Decomposition (SVD) factorizes $A \in \mathbb{R}^{m\times n}$ into $A = U\Sigma V^\top$ [4].

- **Left Singular Vectors ($U \in \mathbb{R}^{m\times m}$):** The columns are the eigenvectors of $AA^\top$. They represent the principal directions in the output space.
- **Right Singular Vectors ($V \in \mathbb{R}^{n\times n}$):** The columns (rows of $V^\top$) are the eigenvectors of $A^\top A$ and represent the principal directions in the input space.
- **Singular Values ($\Sigma \in \mathbb{R}^{m\times n}$):** A diagonal matrix containing non-negative real numbers $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k \geq 0$. These represent the "energy" or variance captured by each respective principal component.

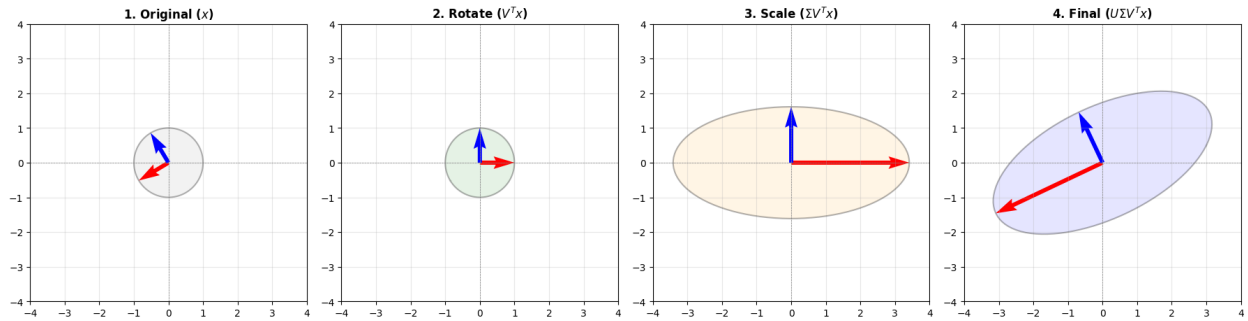


FIGURE 1. The resulting matrices (U, Σ, $V^\top$) from the SVD algorithm represent a translation from the space of the input, for example a vector x, to the output space, Ax. This diagram shows the 'action' of a matrix on vectors in the $\mathbb{R}^2$ unit ball.

2.3. **Randomized Gaussian Sampling.** Multiplying a target matrix A by a Gaussian random matrix Ω (where $\Omega_{ij} \sim \mathcal{N}(0,1)$) functions as a dimension reduction map [2]. This works by ensuring that the sample matrix $Y = A\Omega$ captures the most significant directions of the range of A with very high probability. It acts as a 'sketch' of the operator's range without requiring a computationally expensive full 'drawing'.

2.4. **Theoretical Error Bounds.** The error of the randomized range finder, calculated as $\|A - QQ^\top A\|$, measures how much information is lost by projecting A onto the subspace Q. For a Gaussian test matrix with oversampling parameter p (sampling $k+p$ vectors for a rank-k approximation), the expected error in the spectral norm is bounded by [2]:

$$E\|A - QQ^\top A\| \leq \left(1 + \sqrt{\frac{k}{p-1}}\right)\sigma_{k+1} + \frac{e\sqrt{k+p}}{p}\left(\sum_{j>k}\sigma_j^2\right)^{1/2}$$

The error of the resulting randomized SVD is not identical to the range finder error but is closely coupled. While the range finder establishes the approximation of the subspace, the SVD step computes the best rank-k approximation within that subspace. Per the Eckart-Young theorem, the full SVD error is exactly σ_{k+1}, whereas the randomized version will always be $\geq \sigma_{k+1}$, typically by a small factor determined by the oversampling parameter p [1].

2.5. **Theoretical Performance Bounds.** The computational complexity (Big O) highlights the efficiency of the randomized approach over the deterministic "full" SVD:

Algorithm	**Complexity**
Full SVD	$O(mn \cdot \min(m,n))$
Randomized SVD	$O(mn\log(k) + (m+n)k^2)$

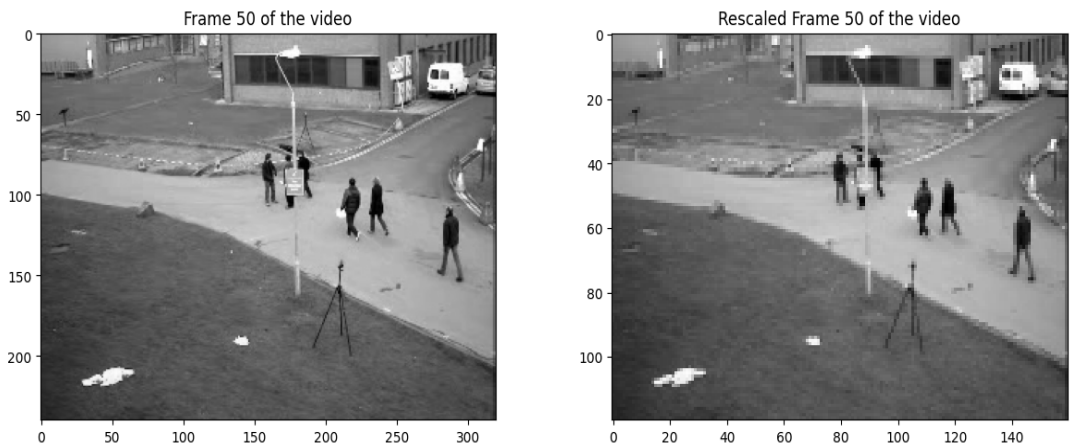
The randomized algorithm transitions the bottleneck from $O(mn^2)$ to $O(mn \log k)$ by utilizing random projections and structured matrix multiplications, making it significantly faster for high-dimensional data where $k \ll n$.

3. Algorithm Implementation and Development

The implementation follows the two-stage probabilistic framework proposed by Halko et al. [2]. The first stage computes an approximate basis for the range of the input matrix, and the second stage uses this basis to perform the decomposition on a significantly smaller matrix.

3.1. Data Import, Rescaling, and Reshaping.

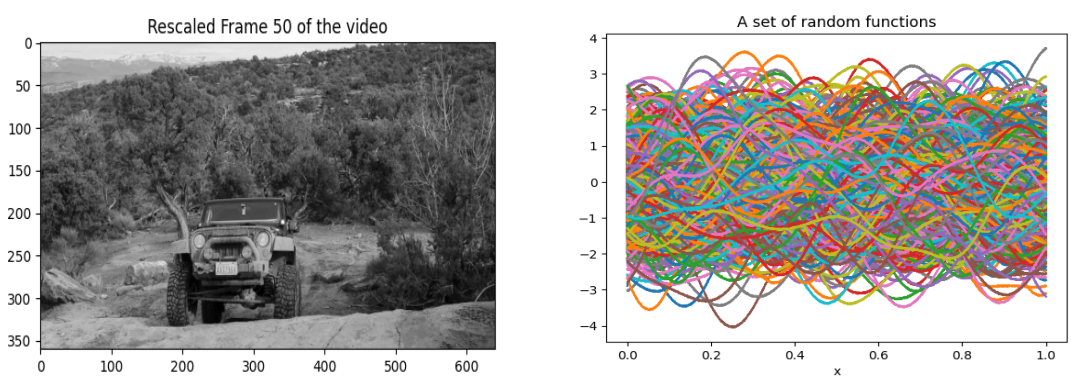
We first import the input data; in this report we compared the performance on a set of random functions, a popular video for testing, and a video from the author's personal library. Video data was converted to grayscale and reshaped into column vectors to form the dense matrix A.



(A) The unaltered video imported at full resolution.

(B) The same frame but scaled down in quality by 0.5.

Figure 2. Frames from the testing video.



(A) A clip from the author's video for further testing.

(B) The baseline test for algorithm development is a set of 1000 randomly distributed functions.

Figure 3. More test datasets.

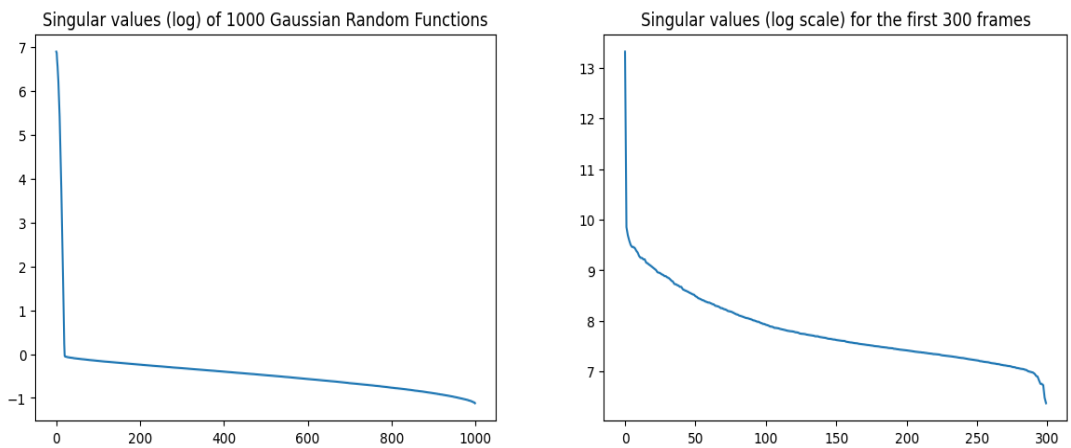
3.2. **Stage A: Randomized Range Finder.** To construct an orthonormal basis Q for the range of A, we first drew a Gaussian random matrix Ω of size $n \times (k+p)$. We then formed the sample matrix $Y = A\Omega$. The QR decomposition was applied to Y such that $Y = QR$, yielding the orthonormal basis Q. This step effectively compresses the information in A into a basis of size $m \times (k+p)$, which is much smaller than n when $k \ll n$ [2].

3.3. **Stage B: Decomposition via Projection.** With the basis Q, we computed the smaller matrix $B = Q^\top A$. This matrix B has dimensions $(k+p) \times n$, allowing for a highly efficient deterministic SVD computation: $B = \tilde{U}\Sigma V^\top$. Finally, the approximate left singular vectors of A were recovered by projecting back to the original space: $U = Q\tilde{U}$. This yields the approximation $A \approx U\Sigma V^\top$ [2].

3.4. **Reshape Result and Output Video.** Finally, the low-rank approximations were reshaped back into the original video dimensions. The low-rank component $(U\Sigma V^\top)$ represents the static background, while the sparse residual $(A - U\Sigma V^\top)$ captures the moving foreground objects.

4. Computational Results

For each dataset, it is important to visualize the singular values to determine whether the SVD, and therefore PCA, is a compatible fit for the use case. We show two of the singular values plotted on a log scale as an example to show our thinking.



(A) Singular values of the random functions. (B) Singular values of the first video.

Figure 4. Two singular value graphs with significant 'energy' held in the first singular values. These are excellent targets for PCA background removal.

By identifying the 'elbow' of these curves, we can choose a k value to slice with and remove the features with the highest frequencies, like the background.

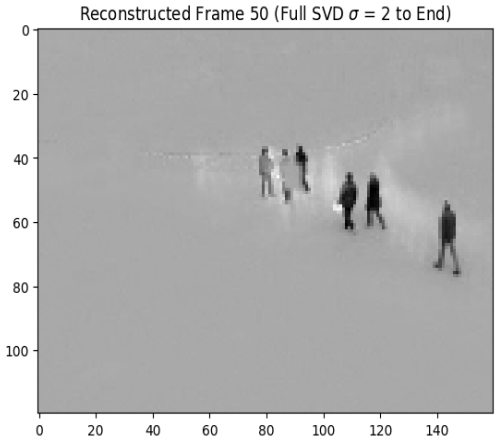
4.1. **Error Results.** Our results show there is significant correlation between the change in variables in the Randomized Range Finder and quantitative results like average and standard deviations. As the parameter k increases, the algorithm keeps more vectors and, therefore, has more capability to encapsulate the range of the data which lowers the average error. When we oversample with p, we see a significant improvement in the standard deviation of our approximation since we are essentially more confident in our results.

Table 1 shows the results for 10 trials on 1000 randomly sampled functions. Note that while E_{SVD} (error relative to full SVD) decreases with p, it is lower bounded by the singular value decay of the matrix itself.

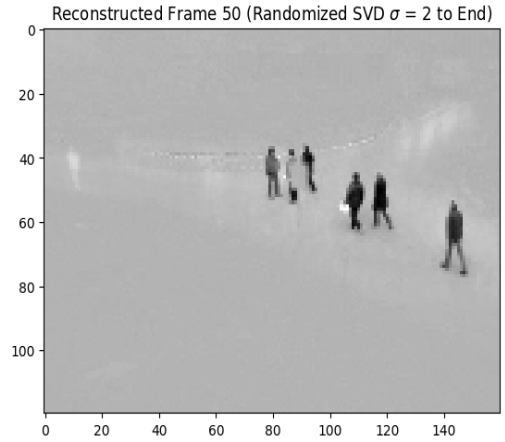
TABLE 1. Comparison of Approximation Errors for Randomized PCA

k	p	$\bar{E}_{\text{mean}}$	σ_{error}	E_{SVD}
5	0	1124.456	96.360	918.049
5	5	797.515	20.529	308.089
5	10	760.016	0.739	40.066
10	0	424.214	63.416	350.739
10	5	162.255	22.495	61.664
10	10	141.500	1.240	26.652
20	0	44.546	3.699	41.311
20	5	33.486	1.173	27.386
20	10	30.514	1.208	20.831

4.2. **Visual and Qualitative Analysis.** Visual inspection of the reconstructed frames (Figures 5 and 6) reveal the rSVD method successfully captures the background structure. Comparing Figures 5a (Full SVD) and 5b (rSVD), the results are visually indistinguishable. This suggests that the randomized approximation error E_{SVD} of 14370.8 (L2 norm difference) consists primarily of high-frequency noise or minor variance that does not impact the dominant background features. In Figure 6b, the "ghosting" artifacts of the moving vehicle are noticable in the low-rank approximation, implying that the motion has not fully been relegated to the residual term.

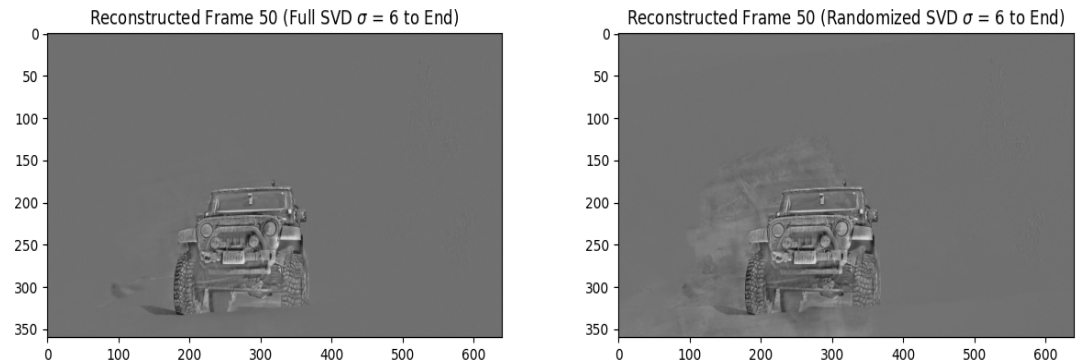


(A) A frame from the full PCA for the rescaled dataset with σ = [2,end].



(B) A frame from randomized PCA for the rescaled dataset with σ = [2,end] and $p = 10$.

FIGURE 5. A performance comparison on the same input dataset between the randomized and full PCA algorithms.



(A) A frame from the full PCA for the entire dataset with σ = [6,end].

(B) A frame from the randomized PCA for the entire dataset with σ = [6,end] and $p = 10$.

FIGURE 6. A performance comparison on the same input dataset between the randomized and full PCA algorithms.

4.3. **Performance Results.** According to the performance bound shown in Section 2.5, we would expect the performance of the Full PCA algorithm to be $O(mn \cdot \min(m, n))$ and the random PCA to be $O(mn \log(k) + (m + n)k^2)$. For the random set of 1000 functions defined at 4096 points, it results in a 4096 x 1000 matrix. Therefore full PCA would be $O(4096 \times 1000 \times 1000)$ and a rank-30 random PCA would be $O(4096 \times 1000 \times \ln(30) + (4096 + 1000) \times 30^2)$, approximately a 200x improvement in computational complexity. Similar improvements were observed during timing of the algorithms for the random functions dataset:

- Average of the full SVD over 10 runs: 6.4355 seconds
- Average of the Randomized SVD (k=30, p=5) over 10 runs: 0.0544 seconds

This represents approximately a 120x speedup.

5. Summary and Conclusions

The comparison between deterministic Full SVD and Randomized SVD confirms that probabilistic algorithms offer a compelling trade-off between speed and accuracy. Our error analysis demonstrates that while the randomized approach introduces an approximation error, this error decays rapidly as the oversampling parameter p increases. Specifically, an oversampling of $p = 10$ was sufficient to bring the error statistics (E_{SVD}) within range for our test cases. Visually, the background subtraction results were indistinguishable between the full and randomized methods. Both successfully isolated the static background (low-rank structure) from the moving foreground (sparse outliers). Given the theoretical complexity advantage—reducing the bottleneck from $O(mn^2)$ to $O(mn \log k)$—the Randomized SVD proves to be a superior choice for large-scale video processing tasks where exact precision is less critical than computational throughput. Future work could explore the impact of power iterations to further refine the singular vectors in cases where the singular value decay is slow [4].

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References

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